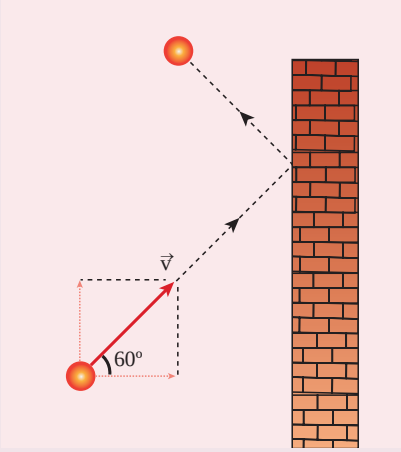
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I. Multiple choice questions

1. A particle of mass m is moving with speed u in a direction which makes 60° with respect to x axis. It undergoes elastic collision with the wall. What is the change in momentum in x and y direction?



(a) $\Delta p_x = -mu, \Delta p_y = 0$

(b) $\Delta p_x = -2mu, \Delta p_y = 0$

(c) $\Delta p_x = 0, \Delta p_y = mu$

(d) $\Delta p_x = mu, \Delta p_y = 0$

2. A sample of ideal gas is at equilibrium. Which of the following quantity is zero?

(a) rms speed

(b) average speed

(c) average velocity

(d) most probable speed

3. An ideal gas is maintained at constant pressure. If the temperature of an ideal gas increases from 100K to 1000K then the rms speed of the gas molecules

(a) increases by 5 times

(b) increases by 10 times

(c) remains same

(d) increases by 7 times

4. Two identically sized rooms A and B are connected by an open door. If the room A is air conditioned such that its temperature is 4° lesser than room B, which room has more air in it?

(a) Room A

(b) Room B

(c) Both room has same air

(d) Cannot be determined

5. The average translational kinetic energy of gas molecules depends on

(a) number of moles and T

(b) only on T

(c) P and T

(d) P only

6. If the internal energy of an ideal gas U and volume V are doubled then the pressure

(a) doubles

(b) remains same

(c) halves

(d) quadruples

7. The ratio $\gamma = \frac{C_p}{C_v}$ for a gas mixture consisting of 8 g of helium and 16 g of oxygen is (Physics Olympiad -2005)

(a) 23/15

(b) 15/23

(c) 27/11

(d) 17/27

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8. A container has one mole of monoatomic ideal gas. Each molecule has f degrees of freedom. What is the ratio of $\gamma = \frac{C_p}{C_v}$